

CLASSIFICATION OF ANEMIA & SIGNIFICANCE OF RED CELL INDICES

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Learning Objectives

At the end of lecture student should be able to

- ▶ Define anemia
- ▶ Describe the classification of anemia
 - Morphological classification
 - Pathophysiological classification
 - Functional classification
- ▶ Enlist the different red cell indices with their definitions.
- ▶ Diagnose anemia by red cell indices

Definition

Anemia (a decrease in the number of RBCs, Hb content, or Hematocrit) below the lower limit of the normal range for the age and sex of the individual.

- ▶ **In adults**, the lower extreme of the normal haemoglobin is taken as 13.0 g/ dl for males and 11.5 g/dl for females.
- ▶ **Newborn infants** have higher haemoglobin level and, therefore, 15 g/dl is taken as the lower limit at birth.

Anemia

Definition:

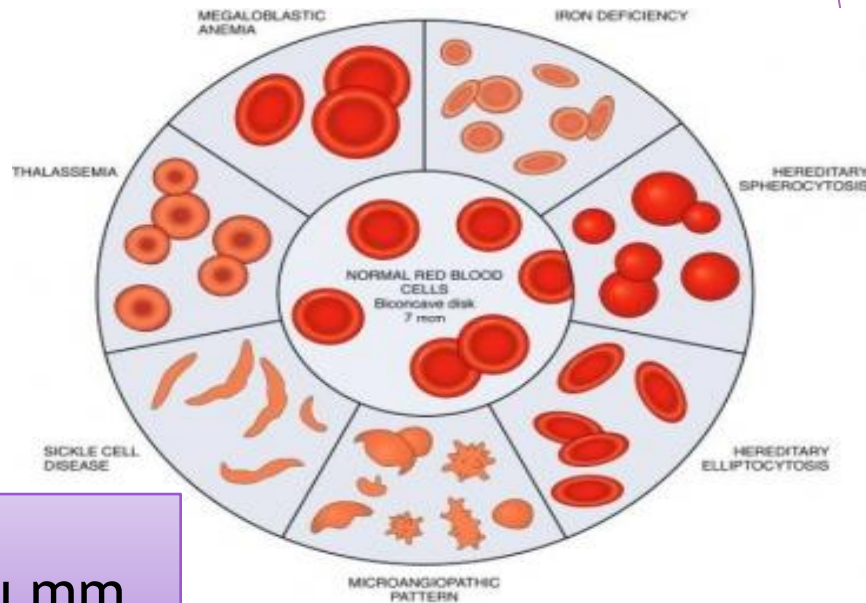
- Decrease in Hb &/or RBC count resulting in decrease O_2 carrying capacity of blood.

Normal Hemoglobin:

- Males □ 14 g/100 ml
- Females □ 12 g/100 ml

Normal RBC count:

- Males □ 5-6 millions/cu mm of blood
- Females □ 4-5 millions/cu mm of blood.



Classification of Anemia

Several types of classifications of anemias have been proposed. Two of the widely accepted classifications are based on

- ▶ The pathophysiology
- ▶ The morphology
- ▶ Functional classification

Morphological classification

Microcyti

C

Normocytic

Macrocyti

C

- Iron def
Anemia
Anemia of
- chronic disease/
Congenital Hb dis.

Hemolytic anemia or blood loss

Anemia of chronic disease
Renal failure
Marrow failure^{low}

Megaloblastic
anemia

Summary of variations in color and size

VARIATIONS IN ERYTHROCYTE COLOR



**NORMOCHROMIC
ERYTHROCYTE**



**HYPOCHROMIC
ERYTHROCYTE**



**HYPERCHROMIC
ERYTHROCYTE**

ABNORMALITIES IN ERYTHROCYTES SIZE



NORMOCYTE



MICROCYTE



MACROCYTE

The morphological classification

Based on red cell size, haemoglobin content and red cell indices anemia are classified into 4 types:

- ▶ Microcytic, hypochromic
- ▶ Normocytic, normochromic
- ▶ Macrocytic, normochromic
- ▶ Macrocytic hypochromic

The pathophysiological classification

Depending upon the pathophysiologic mechanism, anemia are classified into 3 groups:

- ▶ Anemia due to increased blood loss
- ▶ Anemia due to impaired red cell production
- ▶ Anemia due to increased red cell destruction (Haemolytic anaemias)

ACCORDING TO ETIOLOGY

1

↓ Red Cell Production

1. Bone Marrow Disorders (Aplastic Anemia)
2. Chr. Kidney dis. ☐ ↓
Erythropoietin
3. Hypothyroidism
4. Vitamin B12 / Folic acid deficiency
5. Iron deficiency
6. Chronic inflammatory disease

2

↑ Red Cell Destruction

Extracorpascular

1. Erythroblastosis fetalis
2. Transfusion Reactions
3. Autoimmune Hemolytic An
4. Malaria
5. Drugs

3

Blood Loss

1. Acute Blood Loss
2. Chronic Blood Loss

Intracorpascular

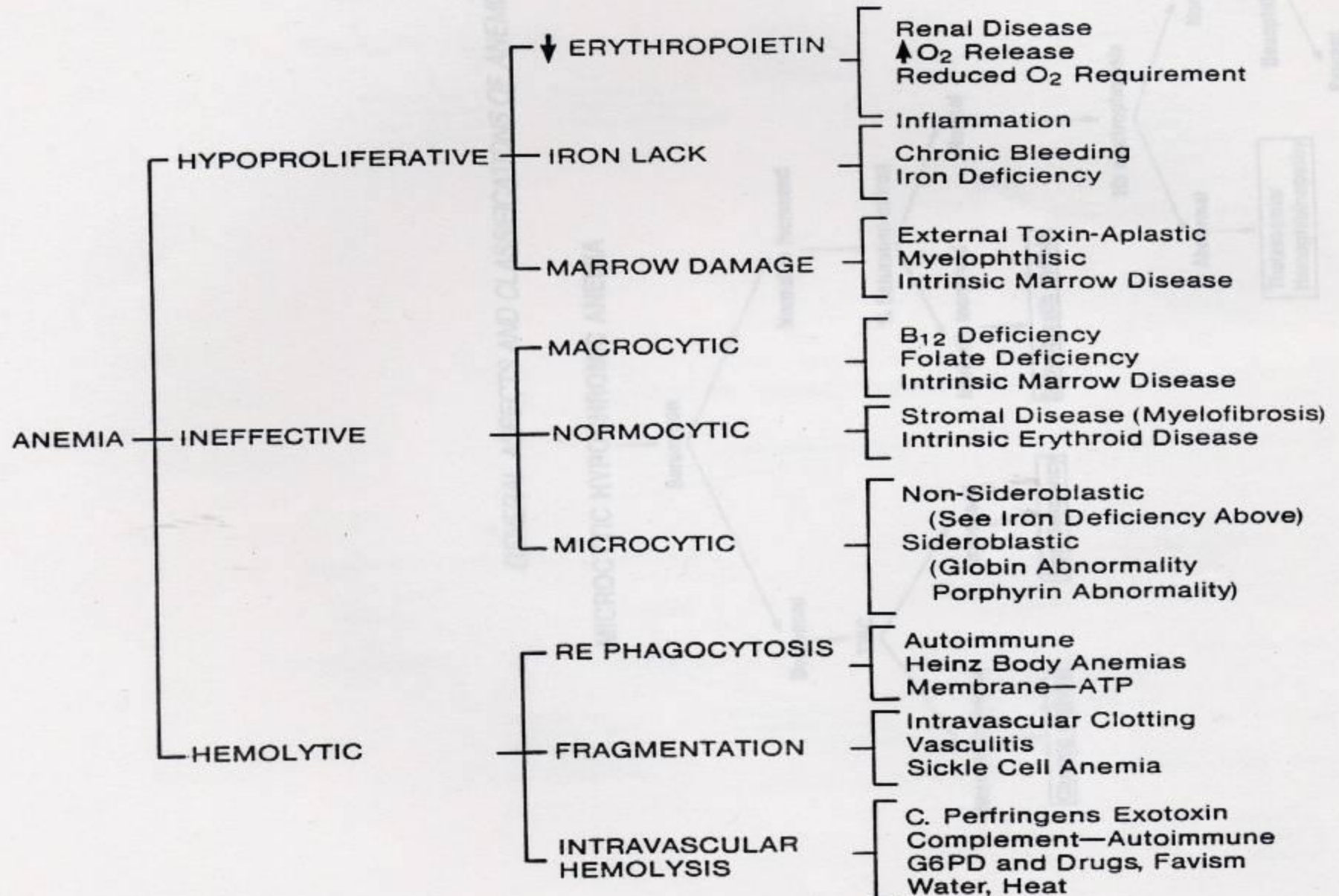
1. Hereditary Spherocytosis
2. Sickle Cell Anemia
3. Thalessemia
4. G6PD deficiency

Functional classification of anemias

Anemias may also be classified functionally into:

- Hypoproliferative (when there is a proliferation defect)
- Ineffective (when there is a maturation defect)
- Hemolytic (when there is a survival defect)

Functional classification of anemias

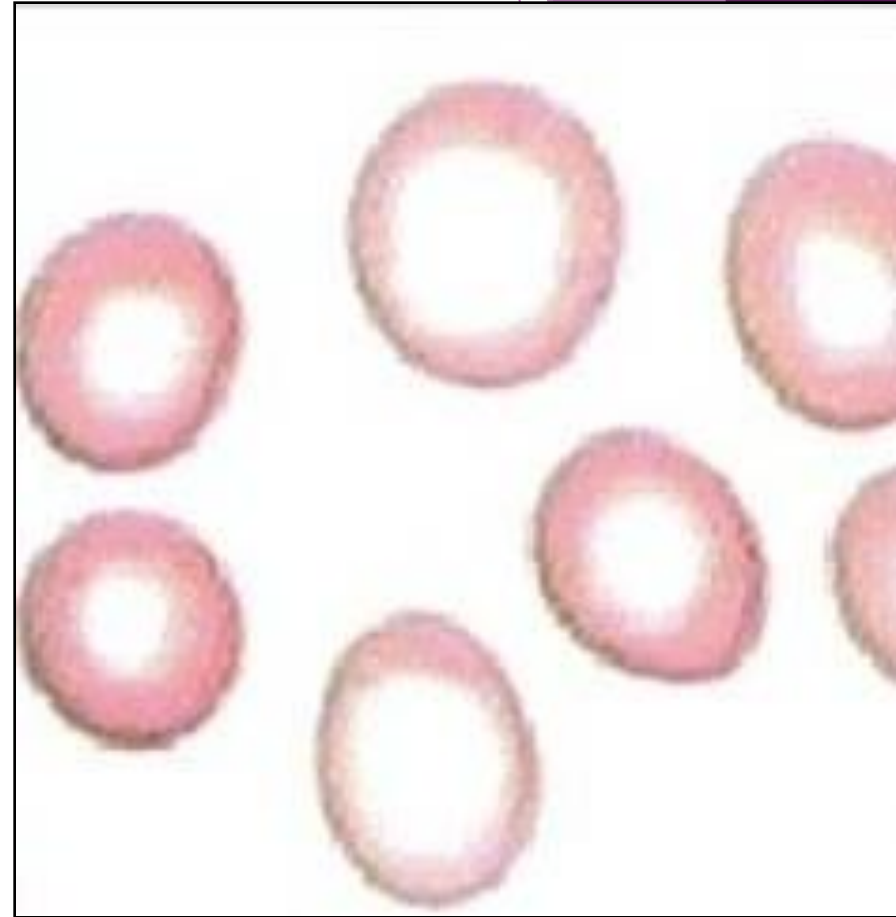
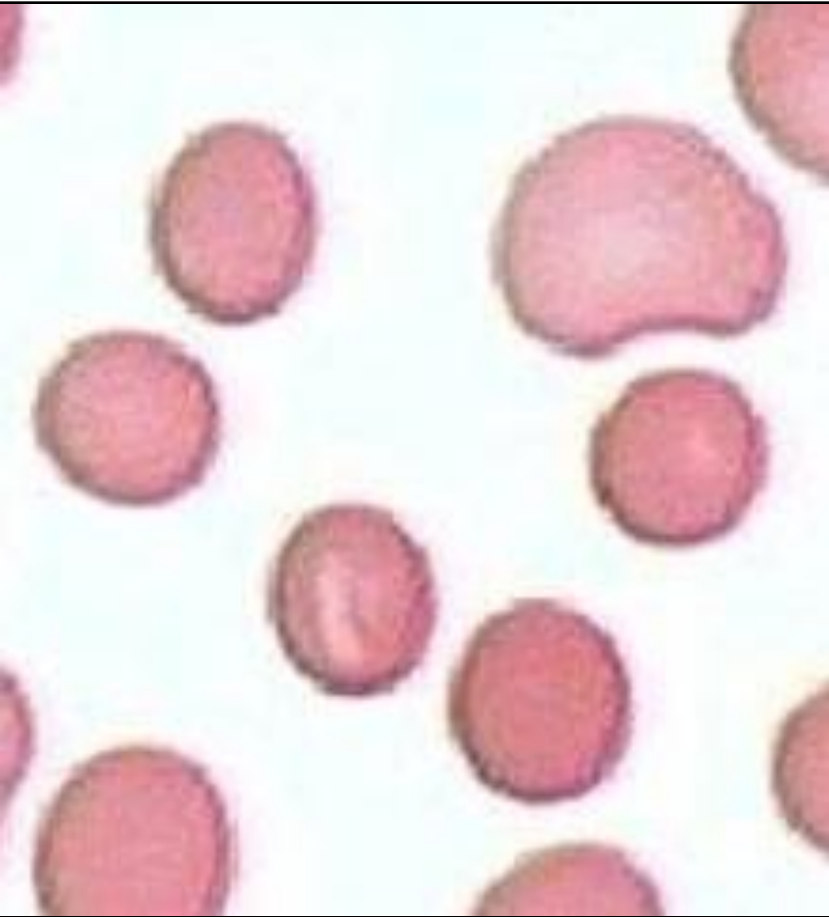


Microcytic, Hypochromic

Causes:

- ▶ Iron deficiency
- ▶ Thalassemia minor
- ▶ Anemia of chronic disease
- ▶ Lead poisoning
- ▶ Congenital sideroblastic anemia
- ▶ β -Thalassemia intermedia and major
- ▶ Hemoglobin H or E disease

Microcytic, Hypochromic cell



Normochromic Anemias

- ▶ Most common cause: Acute blood loss
- ▶ Classifications:
 - ▶ Hereditary Spherocytosis
 - ▶ Hereditary Elliptocytosis
 - ▶ PNH
 - ▶ G6PD deficiency
 - ▶ Aplastic anemia

Normocytic, Normochromic cell



Normocytic, Normochromic

Causes :

- ▶ Anemia of chronic disease
- ▶ Early iron deficiency
- ▶ Renal failure
- ▶ Acquired immunodeficiency syndrome
- ▶ Aplastic anemia
- ▶ Pure red cell aplasia
- ▶ Bone marrow infiltration
- ▶ Leukemia
- ▶ Lymphoma
- ▶ Cancer
- ▶ Granulomatous diseases
- ▶ Myeloproliferative disorder

Macrocytic anemia

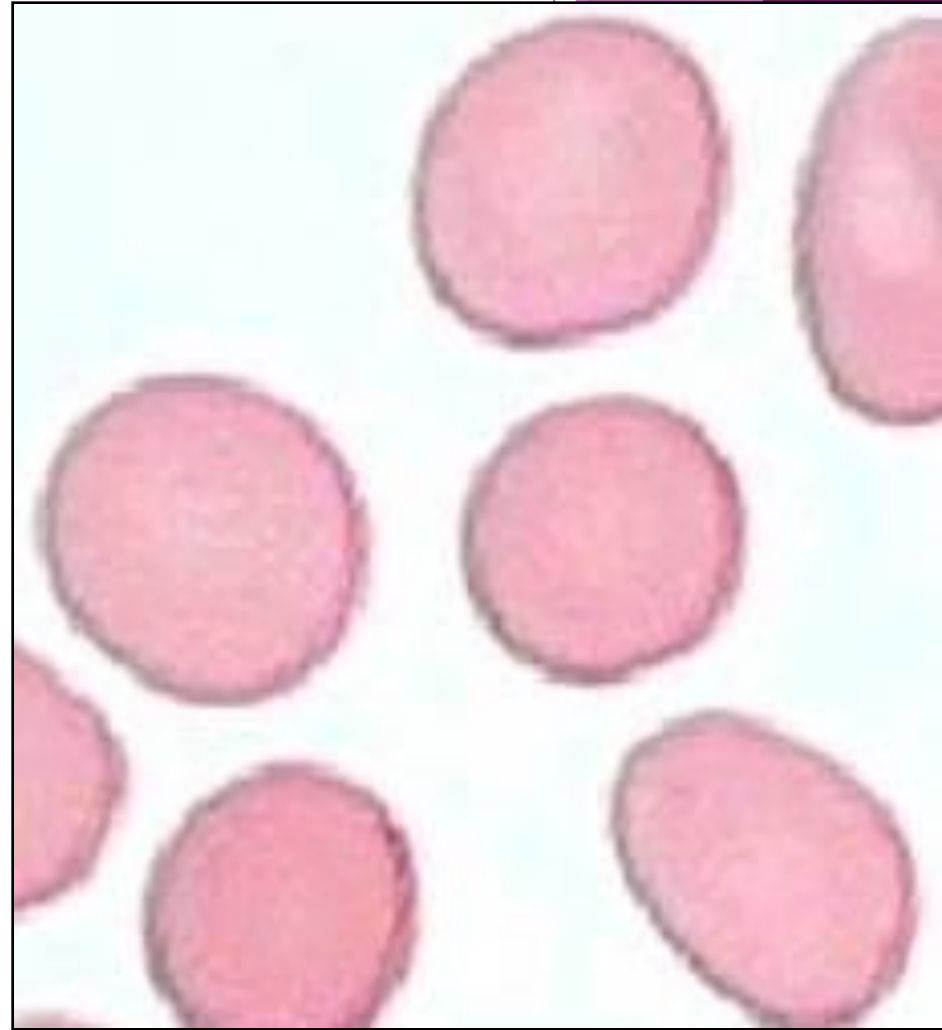
- Vitamin B12 deficiency
- Folate deficiency
- Liver disease

Macrocytic, Normochromic

Causes:

- ▶ Megaloblastic anemia (B12 or folate deficiency)
- ▶ Alcoholism
- ▶ Liver disease
- ▶ Reticulocytosis
- ▶ Chemotherapy
- ▶ Myelodysplastic syndromes
- ▶ Multiple myeloma
- ▶ Hypothyroidism

Macrocytic, Hyperchromic cell

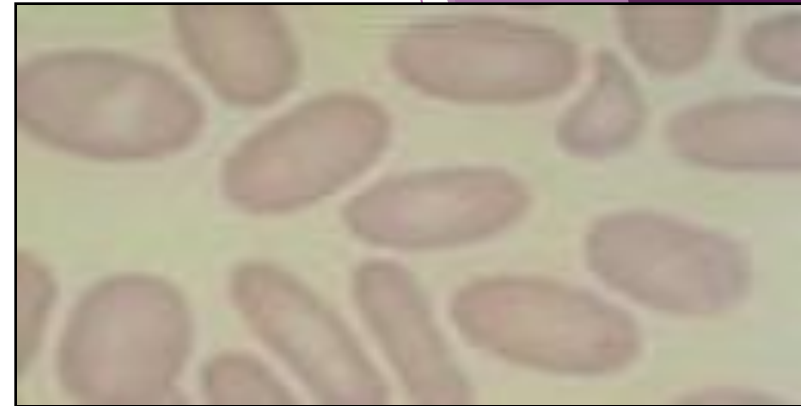


Anaemia; Membrane and enzyme defects

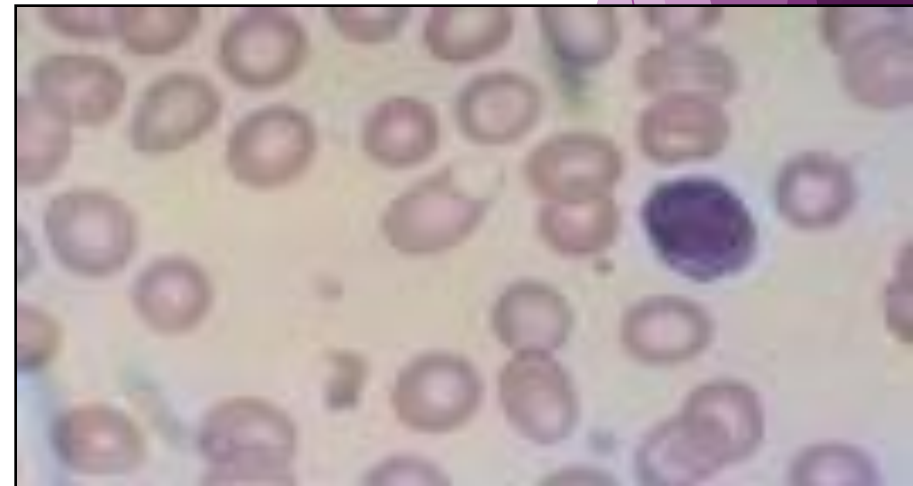
Membrane defects

- Elliptocytosis
 - Hemolysis
- Stomato-ovalocytosis
 - Without haemolysis

Elliptocytosis



G6PD deficiency



Red cell enzymopathies

- G6PD
 - Hemolysis after oxidant stress

Blood loss

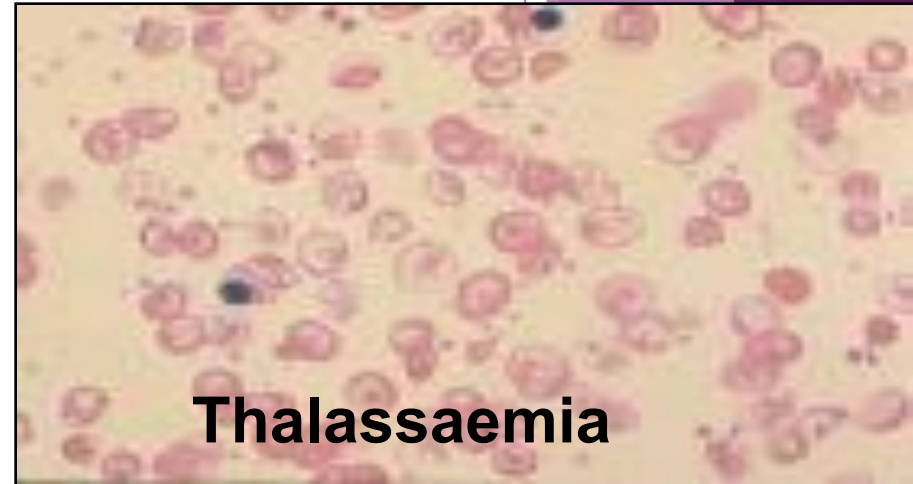
Megaloblastic anemias

- ▶ Are classified morphologically as macrocytic, normochromic.
- ▶ Caused by vitamin B12 & folic acid deficiencies which result in disordered DNA synthesis .

Anaemia; Globin chain defects

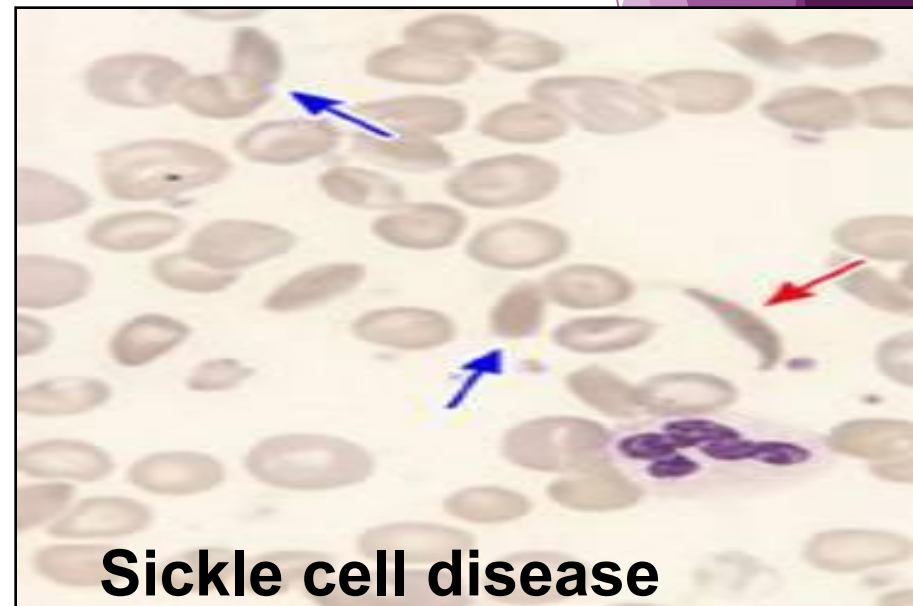
▶ Thassaemias

- ▶ Reduced globin chain synthesis
 - ▶ Alpha and Beta chain synthesis defects



▶ Haemoglobinopathies

- ▶ Abnormal globin chain synthesis



Hemoglobinopathies

Beta Chain Substitutions

Hgb S: Valine for glutamic acid
(6th position, beta chain)

Hgb C: Lysine for glutamic acid
(6th position, beta chain)

Laboratory Investigation

- ▶ Anemia is not a diagnosis, but a sign of underlying disease.
- ▶ The objective of the laboratory is to determine the type of anemia as an aid in discovering the cause.

RBC INDEX



- It is a quantitative measurement of red blood cell, the amount and concentration of hemoglobin in them.
- They were first introduced by Wintrobe in 1929 to define the size (MCV) and hemoglobin content (MCH & MCHC) of red blood cells

Red cell indices

- ▶ Mean cell volume (MCV)
- ▶ Mean cell hemoglobin (MCH)
- ▶ Mean cell hemoglobin conc. (MCHC)

DEFINITIONS:



- **Mean cell volume (MCV)**
 - It is the measure of average volume of RBCs
- **Mean cell Hemoglobin (MCH)**
 - It is a measurement of the average weight of hemoglobin in individual erythrocytes.
- **Mean Cell Hemoglobin Concentration (MCHC)**
 - It is the average concentration of hemoglobin in erythrocytes
- **Red cell distribution width (RDW)**
 - It is a measure of variability of erythrocyte size

Mean cell volume (MCV)

- ▶ Normal reference range 76 fl - 100 fl
- ▶ Is used to differentiate between types of anemia on the basis of red cell size
- ▶ If > 100 fl , then macrocytic anemia
- ▶ If < 76 fl , then microcytic anemia
- ▶ Calculated as Hematocrit / RBC count

Mean cell hemoglobin (MCH)

- ▶ Defined as average mass of hemoglobin per red blood cell in a sample of blood
- ▶ Normal reference range is 27 - 31 pg /cell
- ▶ The value decreases in hypochromic anemias and increases in hyperchromic anemias

Mean cell hemoglobin conc. (MCHC)

- ▶ Defined as concentration of [hemoglobin](#) in a given volume of packed [red blood cell](#)
- ▶ Normal reference range is 32 - 36 g/dl
- ▶ Calculated as hemoglobin / hematocrit
- ▶ This count is used to give a rough guide to what shade of red the RBC will be. (paler=lower than the standard)

Nutritional deficiency anemia clinical application

Angular Cheilosis



Koilonychia



Marrow iron stores



Table 2.1: Normal range for red cell and other parameters

	<i>At birth</i>	<i>Men</i>	<i>Women</i>
Red cell count	5 to 7 million/cu mm ($5-7 \times 10^{12}/L$)	4.5 to 5.5 million/cu mm ($4.5-5.5 \times 10^{12}/L$)	3.8 to 4.8 million/cu mm ($3.8-4.8 \times 10^{12}/L$)
Hemoglobin (Hb)	14–22 g/dL	13–17 g/dL	12–15 g/dL
Hematocrit (Hct) or packed cell volume (PCV)	45–70%	40–50%	36–46%
Reticulocyte count	1–7%	0.5–2.5%	0.5–2.5%
Red cell indices			
MCV	100–120 fL	82–100 fL	82–100 fL
MCH	31–37 pg	27–32 pg	27–32 pg
MCHC	30–36 g/dL	31–35 g/dL	31–35 g/dL
RDW	13–18%	11.5–14.0%	11.5–14.0%

References

- ▶ Medical physiology
 - ▶ Guyton & Hall
 - ▶ 13th edition
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- ▶ Medical physiology
 - ▶ Ganong
 - ▶ 23rd edition

Thank
You!!

